Random Forest Algorithm for Alzheimer's Disease Prediction

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Abstract. Alzheimer's disease is a gradient degeneration of essential cognitive activities that mainly affects elderly individuals. Diagnosis of Alzheimer's disease by neuropsychological assessments is considered an important step in disease management. However, using a single neuropsychological assessment technique often leads to a high rate of misdiagnosis of the disease. To tackle this issue, we built a Random Forest machine learning algorithm with five different neuropsychological assessments to predict Alzheimer's disease. A total of 1761 samples acquired from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database were classified into cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer's disease groups. we create three separate subsets, namely CN vs. AD, CN vs. MCI, and AD vs. MCI for train and test the Random Forest model. Our model achieved high accuracy in distinguishing between the healthy and affected groups, it attained an accuracy of 97.74% with the CN vs. AD subset and 94.65% with the CN vs. MCI subset, making it capable of quickly diagnosis of AD. Our study suggests a robust and efficient Random Forest classifier for Alzheimer's disease prediction.

Keywords: Alzheimer's disease, Dementia, Random Forest, Neuropsychological Assessment.

1 Introduction

Alzheimer's disease (AD) is a gradient degeneration of essential cognitive activities such as memory, thinking and cognition, and mainly affects people who are 60 years or older. Generally, there are three different stages of Alzheimer's disease: early-stage, middle-stage, and late-stage.In the early stage, patients begin to show simple and progressive loss of cognition and memory, with symptoms such as mild forgetfulness and problems with concentration. In the middle stage, as the disease progresses, patients experience trouble remembering events, learning new things, and planning complicated events. In the late stage, patients lose some physical abilities, such as walking, sitting, eating, and speaking [1], [2]. Several neuropsychological assessments are used for cognition measures such as Geriatric Depression Scale (GDSCALE) which evaluates symptoms of depression in elderly patients. Mini-Mental State Exam (MMSE) that measures short- and long-term memory; Global Clinical Dementia Rating (GCDR) that assesses cognitive and functional abilities; Functional Activities Questionnaire (FAQ)

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that is a clinical-report questionnaire that measures instrumental activities of daily living. and Neuropsychiatric Inventory Questionnaire (NPI-Q) that is used to assess a range of symptoms in individuals with dementia such as agitation, depression, delusions, hallucinations, and sleep disturbances. These neuropsychological questionnaires are commonly used to assess different levels of cognition and are frequently used to diagnose AD [3],[4].

Major of recent studies have primarily focused on analysing neuroimaging biomarkers or the combination of neuroimaging with other biomarkers, which increases the cost of diagnosis and limits the ability to generalize the diagnostic process in the first-line.

The objective of this study is to develop a Random Forest machine learning model for accurately predicting Alzheimer's disease at a minimal expense, to help patients with AD receive timely clinical intervention and avoid delays in diagnosis.

2 Materials and Methods

2.1 Dataset Description

First The neuropsychological data used in this study is acquired from Alzheimer's Disease Neuroimaging Initiative database ADNI (http:// adni.loni.usc.edu). ADNI provide unlimited access to the database and encourages researchers to use its data for understanding Alzheimer's progression.

In this study, we used data collected during Phase 1 of the ADNI (ADNI-1). The initial dataset contained numerous empty and unnecessary fields, which were cleaned by removing all unnecessary and empty columns, and the non-numeric values were replaced with numeric values. The final dataset consisted of 1761 samples, categorized into three groups: 616 Cognitively Normal (CN), 878 Mild Cognitive Impairment (MCI), and 267 Alzheimer's disease (AD). In addition to the age and gender information, the dataset included five neuropsychological assessments conducted at different time intervals, namely GDSCALE, MMSE, GCDR, FAQ, and NPI-Q. Table 1 presents the information about the subjects.

Table 1. Demographics and information of subjects.

Clinical Group	Sample Size	Avr. Age	Female
CN	616	77.96	290
MCI	878	76.72	300
AD	267	77.07	127

2.2 Random Forest Algorithm (RF)

Random Forest is an ensemble learning algorithm that uses multiple decision trees and combines them to improve the prediction performance. A decision tree is a hierarchical machine learning model composed of decision rules, that are applied to partition the dataset feature space into subspecies of one class. Building a decision tree

classifier is based on recursively partitioning the feature space of the training set to obtain an optimal set of decision rules that provide the best classification model. Random Forest algorithm constructs each decision tree in the ensemble using a bootstrapped sample of the training dataset, where each sample has an equal chance in selection. Additionally, the ensembles are constructed using a sample of data that is randomly selected from the training dataset with replacement every time. Each tree in the ensemble acts as an autonomous classifier to determine the class label of an unlabelled instance using an aggregating technique called majority voting, where each classifier casts one vote for its predicted class label, and then the class label with the most votes is used to classify the unlabelled instance. [6], [7]. Fig 1. presents the block diagram of the Random forest algorithm.

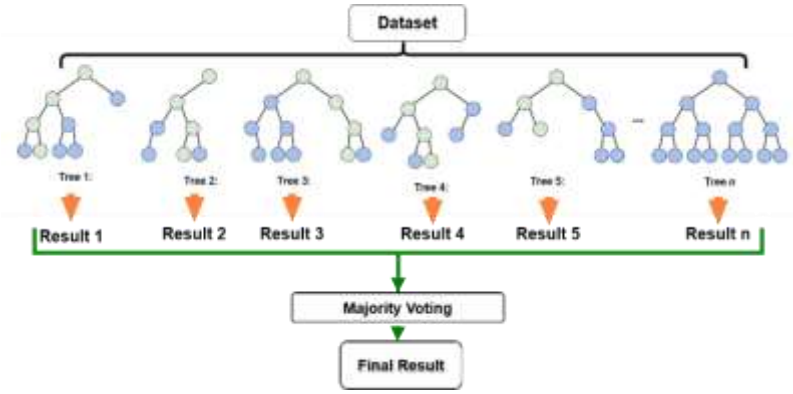


Fig. 1. Block diagram of random forest algorithm with n decision trees.

2.3 Proposed Framework

The framework used for model development is based on two steps: The first step entails training the Random Forest (RF) model using various hyper-parameter configurations to identify the optimal hyper-parameters. To achieve this, the GridSearchCV tuner from Scikit-learn library, with 5-fold cross-validation, was employed. GridSearchCV explores a predefined range of hyper-parameters and exhaustively searches through all possible combinations to determine the best combination that yields the highest performance accuracy for the model. The second step involves retraining the model using the optimal hyper-parameters obtained from the first step. Fig.2. presents the block diagram of the proposed framework.

The final dataset was divided into three subsets: NCvsAD, NCvsMCI, and ADvsMCI. In each time, 80% of each subset was used for training the Random Forest model, while the remaining 20% was reserved for testing. The hyper-parameters that were tuned included max_depth, max_features, and max_leaf_nodes. The optimal configuration for the model was selected, and the model was retrained and evaluated using a five-fold of cross-validation algorithm.

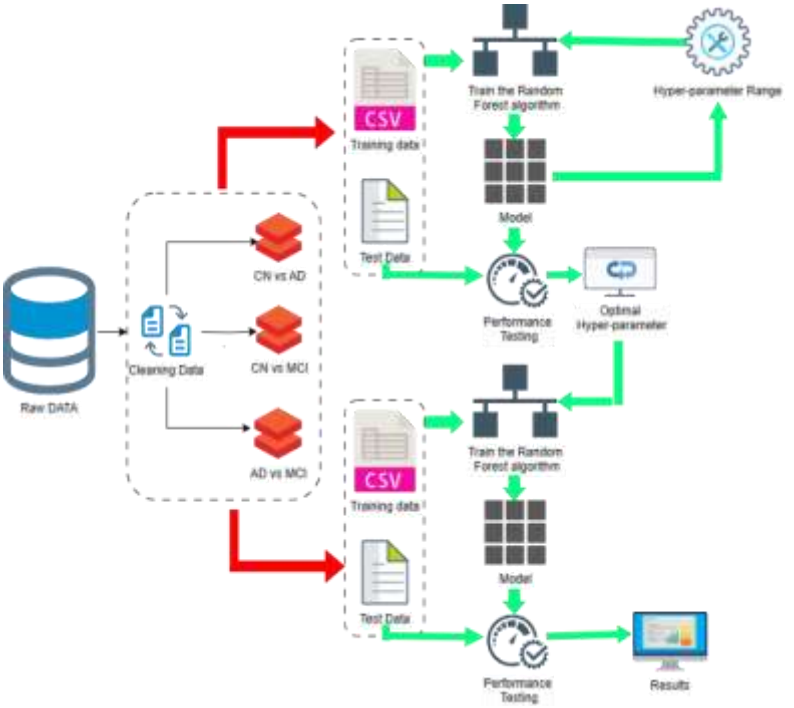


Fig. 2. Block diagram of the proposed framework.

3 Results

In this study, 1761 subjects were classified into three categories: CN, MCI, and AD. The classification was performed using the One-vs-One classification strategy with the Random Forest algorithm.

3.1 Hyper-parameter Optimization

To determine the optimal hyper-parameters of the RF model Max_depth, max features, max_leaf_nodes, and the cost function parameters were tuned. The hyper-parameters that yielded the highest accuracy score were selected as optimal. Table 2 shows the optimal hyper-parameter selected.

Table 2. Optimal hyper-parameters selected

Hyper-parameter	Hyper-parameter range	Optimal hyper-parameter		
		CNvs AD	CNvs MCI	ADvs MCI
Max_depth	4 to15	10	8	12
Max_features	'auto', 'sqrt', 'log2'	sqrt2	Auto	log2
N_estimators	200, 500,1000	200	200	200
Cost function	'Gini', 'Entropy'	Entropy	Gini	Entropy

3.2 Performance results

Initially, we split each subset into two essential parts: training set that contain 80% of the total data, which was used to train the RF classifier, and test set that contain 20% of the total data, which was used to evaluate the performance of the trained model. The results of the overall accuracies for each subset and Area Under the Curve (AUC) are presented in Table 3, and for more detailed performance results, Tables 4 summarize the performance of the model for the CNvsAD, CNvsMCI, and ADvsMCI subsets, respectively. Table 3 provides also an additional insight into the classification performance for each specific category. Furthermore, Fig. 2. displays the Receiver Operating Characteristic (ROC) curves, providing a visual representation of the trade-off between the true positive rate and false positive rate of the model. These curves are helpful in evaluating the ability of the model to distinguish between positive and negative instances. Moreover, the area under the ROC curve (AUC) is calculated, which represents the probability of correctly classifying a positive instance higher than a negative instance.

Table 3. Overall accuracy and AUC of the subsets

Subset	Accuracy	AUC
CN vs AD	97.74%	0.98
CN vs MCI	94.65%	0.93
AD vs MCI	80.35%	0.69

Table 4. Performance Metrics of the model

Subset	Accuracy	Precision	Recall	F1 score
CN vs AD	97.74%	95.92%	98.48%	97.10%
CN vs MCI	94.65%	95.00%	93.95%	94.40%
AD vs MCI	80.35%	75.57%	68.50%	70.57%

The experimental results showed that the RF model achieved the highest accuracy 97.74% with the subset CNvsAD followed by 94.64% accuracy for CNvsMCI subset, while the RF model showed the lowest accuracy 80.35% for ADvsMCI subset.

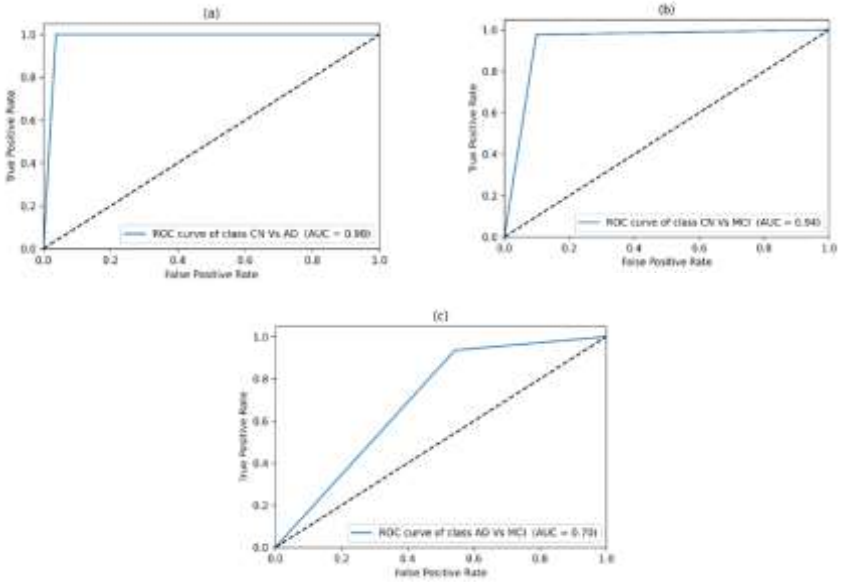


Fig. 3. Receiver Operating Characteristic (ROC) curves of the three subsets. (a) ROC curve of CN-vs-AD subset. (b) ROC curve of CN-vs-MCI subset. (c) ROC curve of AD-vs-MCI subset.

4 Discussion

In this study, we successfully implemented and fitted a Random Forest model to classify individuals with AD dementia into three stages using the one-vs-one strategy based on neuropsychological data. The findings indicate that the performance of any machine learning model is influenced by the correct selection of hyper-parameters associated with it. Therefore, choosing the most appropriate hyper-parameters is crucial for achieving optimal performance, as several studies have shown the effectiveness of this technique in improving model performance [8]. In our study, we employed the GridSearchCV tuner to adjust and optimize the hyper-parameters of the model, intending to identify the hyper-parameters that would yield the best performance for our classification task. However, it is important to note that there is no definitive rule for choosing the best hyper-parameters, as their optimal values can vary depending on the dataset being used. Each dataset has its own unique characteristics, and the performance of hyper-parameters can be influenced by these specific attributes. Therefore, the selection of hyper-parameters should be based on careful experimentation and validation using appropriate techniques such as cross-validation. While GridSearchCV is a widely used technique for hyper-parameter tuning, it does have some limitations, one of the main drawbacks is that it can be very time-consuming, as it needs to test every possible combination of hyper-parameters. This can be a challenge when dealing with large datasets or complex models, requiring significant computational power and time. However, there are alternative optimization techniques available that can help address this issue. For example, RandomizedSearchCV and HalvingGridSearchCV from Scikit-

learn can test a broader range of hyper-parameter values within the same computation time as GridSearchCV, making them more efficient options for hyper-parameter tuning.

Our results also showed that the random forest classifier model achieved the best accuracy (97.74%) in the CNvsAD subset. This indicates that it is highly appropriate for distinguishing between individuals diagnosed with AD dementia and those who are not diagnosed, making the model robust in predicting AD. Similarly, the CNvsMCI subset yielded 94.65% as accuracy, that demonstrate that the MCI individuals had a notable neuropsychological disorders compared to the healthy individuals because the classifier was able to distinguish between them. On the other hand, the ADvsMCI subset yielded a low classification accuracy (80.35%) because the MCI subjects are individuals who are likely in the early stages of AD, and their neuropsychological scores do not directly reflect the disease, leading to poor classification performance. This limitation prompts to consider in future work to combine neuroimaging and neuropsychological data for predicting the early stage of Alzheimer's disease. However, our proposed model can distinguish between both Alzheimer's disease patients and individuals who are likely affected by the disease (MCI) on one hand, and healthy individuals on the other.

The findings of the study reveal that the AUC value is 0.98 for the CNvsCN group and 0.94 for the CNvsMCI group. These high AUC values suggest that the model has a strong ability to correctly classify positive instances. This implies that the model performs well in distinguishing healthy individuals (CN) from those not (AD and MCI). Based on our findings, the classification performance of our model is comparable to previous studies that utilized traditional machine learning algorithms and neuropsychological data. Our model achieved the highest accuracy compared to several previous studies. For instance, Khan Afreen et al. reported a best accuracy of 93.90% using their RF model [9]. Lemos Luís et al. developed an ML model to discriminate between AD and MCI, with a best accuracy of 86% [10]. Ibrahim Almubark et al. pointed out that traditional machine learning algorithms attained similar classification performances when utilizing neuropsychological data, and when they used neuropsychological scores, their MLP model achieved an accuracy of 84.21% [11].

It is essential to interpret our findings considering several limitations. Firstly, the results of the research may have been influenced by missing data, potential coding errors, and unmeasured factors, such as neuropsychological variables and other clinical variables. Additionally, other diseases such as schizophrenia and Parkinson's disease can exhibit similar psychological symptoms to AD, which could lead to difficulties in distinguishing between them and AD. Furthermore, the limited number of subjects in the study highlights the need for future research to enhance the dataset by incorporating other markers, such as brain imaging, genetic and biological markers.

5 Conclusion

In this paper, we have developed a robust Random Forest classifier for predicting Alzheimer's disease based on neuropsychological data. We used five common neuropsychological assessments to train the model. The findings demonstrate that the model achieved a high classification performance in distinguishing between the Healthy

group (CN) and the affected groups (AD and MCI). This study presents a robust machine learning model capable of accurately predicting Alzheimer's disease at minimal expense. It is recommended to use this model as a first-line diagnostic tool for Alzheimer's disease. The future work involves combining the neuropsychological marker with various biological and imaging markers to develop a robust machine learning model capable of predicting the early stages of AD.

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Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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